

FIFA 21 Player Data Analysis: A Comprehensive Exploration Using Python and Streamlit

Abstract

The increasing availability of data in the world of football has revolutionized how players are evaluated, teams are formed, and strategies are developed. This study presents an in-depth analysis of the FIFA 21 player dataset using a combination of Python-based analytical tools and Streamlit for creating an interactive web application. The objective is to transform raw player statistics into meaningful visual insights that can support data-driven decisions for scouts, analysts, and fans. Our approach includes preprocessing of the data, exploratory data analysis (EDA), and the development of a feature-rich dashboard. The dashboard is divided into various sections such as Home, Player Comparison, Team Analysis, and Feature Distributions. It highlights player market value, wage structure, performance attributes, and national trends. This paper elaborates on each of these sections, discussing the methodology, visual outputs, and the analytical value they offer. We conclude by outlining future work and potential improvements.

1. Introduction

Data analytics has transformed many domains, and professional football is no exception. From scouting talent to optimizing player performance and developing game strategies, the insights derived from data play a vital role. The FIFA 21 video game dataset is an excellent proxy for real-world player statistics. Though simulated, it closely mirrors real-life assessments and valuations of players across various leagues and countries. This study leverages the FIFA 21 dataset to analyze over 18,000 football players using statistical and visualization techniques.

The primary goal is to convert a high-dimensional dataset into interactive, digestible insights through a Streamlit dashboard. By creating intuitive visuals like radar charts, histograms, bubble plots, and boxplots, we aim to make player evaluation accessible not just to data scientists but also to football enthusiasts and professionals.

2. Dataset Overview

The FIFA 21 player dataset includes a diverse array of variables:

- **Personal Information:** Name, date of birth, nationality, club affiliation, and preferred foot.
- **Performance Metrics:** Pace, shooting, passing, dribbling, defending, and physical attributes.

- **Market Metrics:** Overall rating, potential rating, player value in euros, and weekly wage.
- **Goalkeeping Attributes:** Specific to goalkeepers, such as diving, reflexes, handling, etc.

2.1 Data Preprocessing

- Date columns such as `dob` (date of birth) and `joined` were converted to datetime format.
- Player age was calculated based on the year 2020.
- `value_eur` and `wage_eur` were normalized to millions of euros for easier interpretation.
- Column names were cleaned and standardized for clarity.

After cleaning, the dataset was ready for exploration and dashboard integration.

3. Methodology

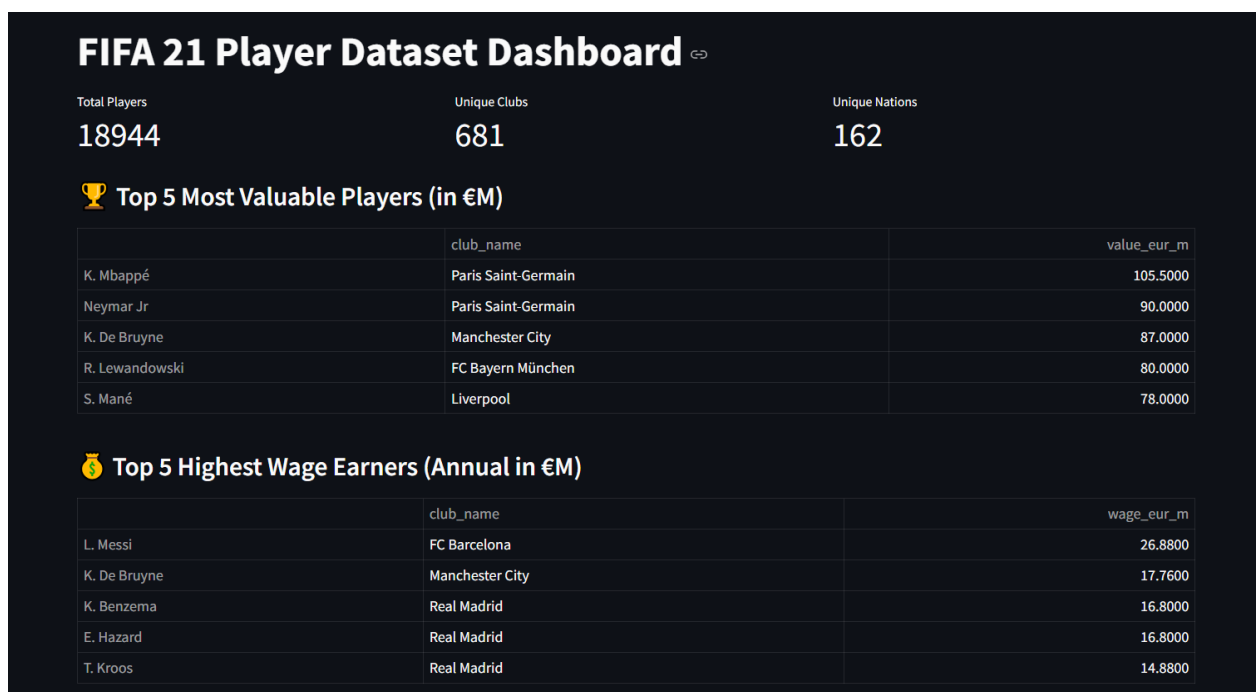
3.1 Tools and Technologies Used

- **Python:** Core scripting and data processing language.
- **pandas:** Used for loading and manipulating data.
- **seaborn & matplotlib:** For generating various visualizations.
- **Streamlit:** To build an interactive web dashboard that users can explore in real time.

3.2 Dashboard Architecture

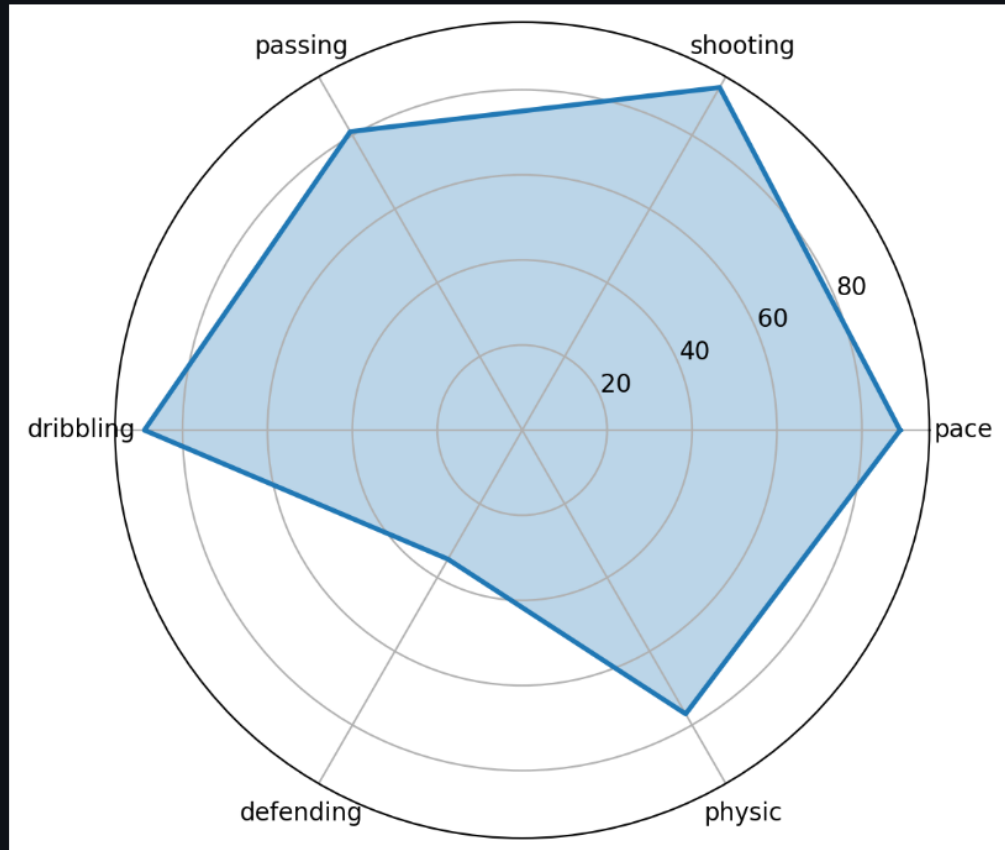
The dashboard is divided into four primary sections:

- **Home Page:** Overview metrics, top-valued players, wage earners, age distribution, and value vs. overall rating bubble chart.

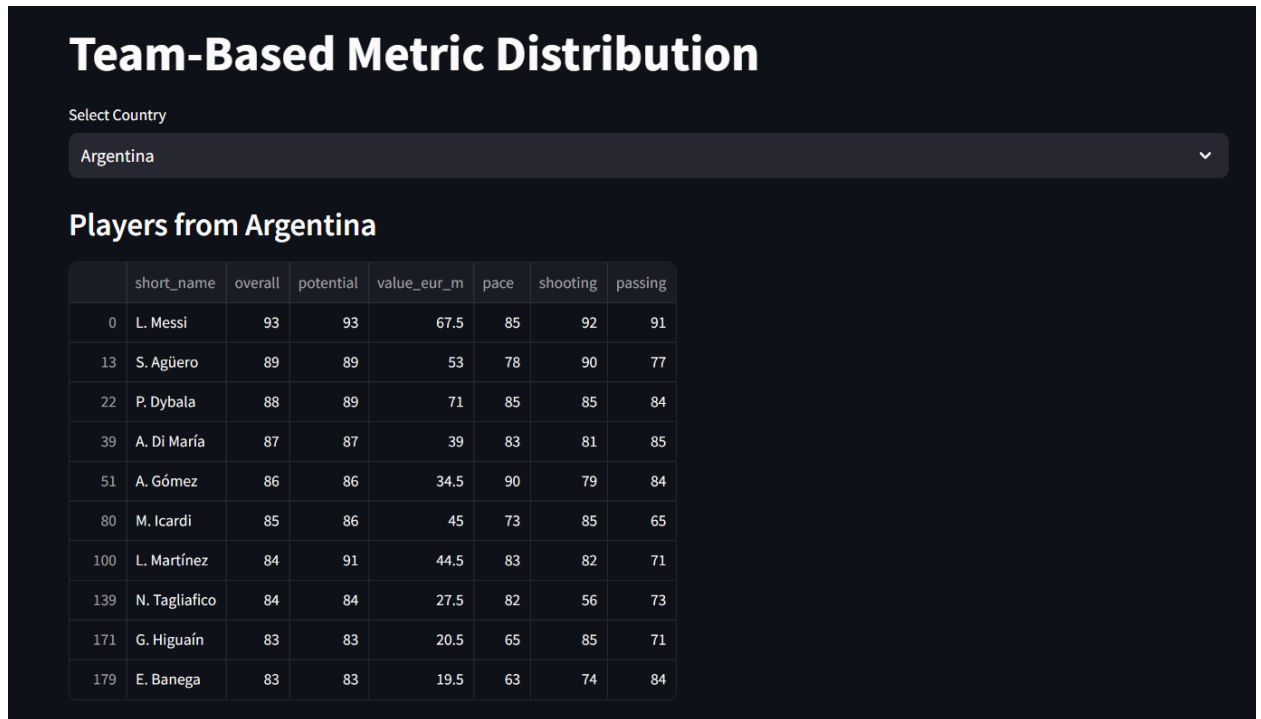


- **Player Comparison:** Radar chart comparisons for selected players based on core performance metrics.

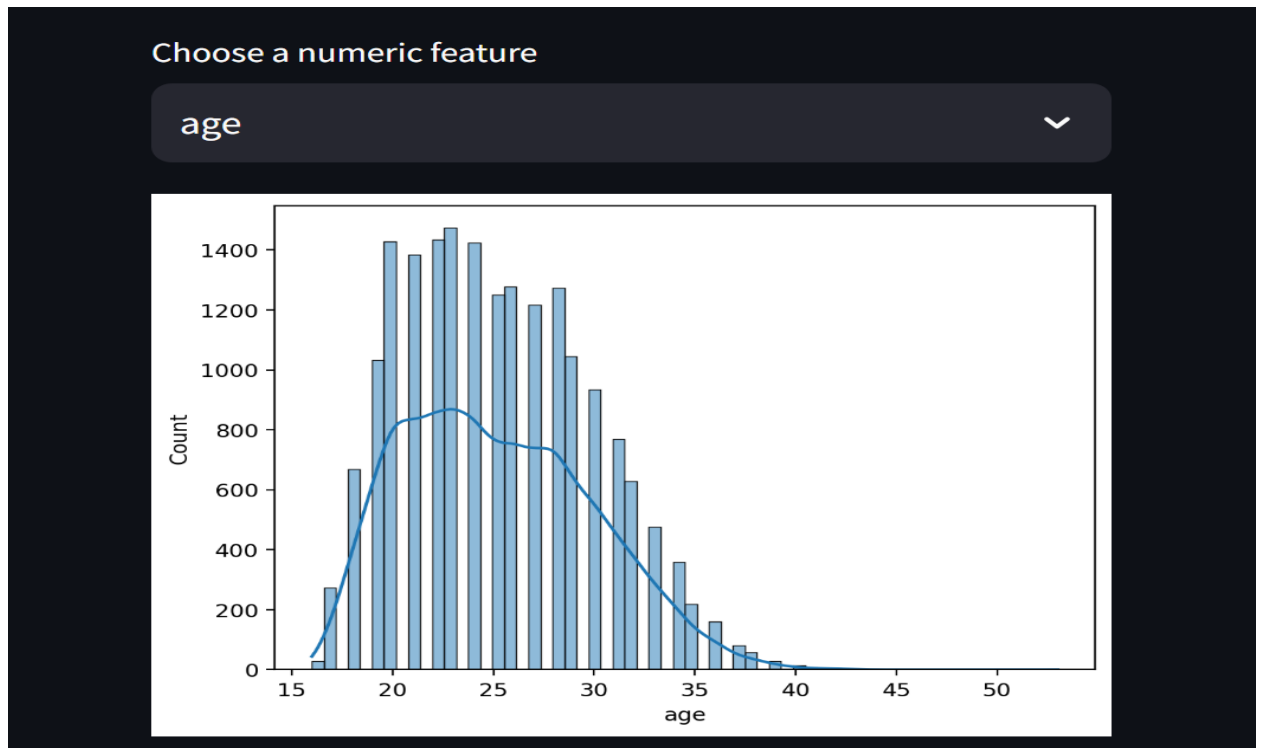
Cristiano Ronaldo



- **Team Analysis:** Filters players based on nationality and shows statistical distributions of selected metrics.



- **Distributions:** Enables selection of a numerical feature to visualize its distribution through histograms and boxplots.



4. Results and Analysis

4.1 Home Section The home page delivers an immediate overview of the dataset:

- **Total Players:** ~18,000
- **Unique Clubs:** 698
- **Unique Nations:** Over 150

Top 5 Most Valuable Players:

- Includes iconic figures like Lionel Messi, Cristiano Ronaldo, and Neymar.
- Sorted by value in millions of euros.

Top 5 Wage Earners:

- Highlights market wage disparity.
- Suggests correlation between club reputation and wage offerings.

Age Distribution:

- Displays a right-skewed distribution.
- Majority of players are between 20 and 28 years old.

Bubble Chart (Overall vs. Value):

- Uses bubble size to represent player wage.
- Confirms a positive correlation between overall rating and market value.

4.2 Player Radar Comparison This section allows users to select multiple players and compare them across six core metrics:

- **Pace, Shooting, Passing, Dribbling, Defending, Physic**

Radar charts visually emphasize each player's strengths and weaknesses. For example:

- Messi excels in dribbling and passing.
- Ronaldo stands out in shooting and physical attributes.

The radar chart format makes it easy to see how players complement each other, useful for team formation.

4.3 Team Analysis Section Users can select a country to filter and analyze players of that nationality.

- Shows top 10 players by overall rating, value, and core metrics.
- Generates a histogram for a selected attribute (e.g., pace for Brazilian players).

This helps identify:

- Countries with higher average pace or technical skill.
- National football identities (e.g., strong physical players from African nations).

4.4 Distributions Section This section offers flexible feature exploration:

- Users can pick any numeric feature.
- Histograms show the spread and skewness.
- Boxplots highlight outliers and median values.

This is especially helpful for:

- Understanding outliers in wage or value.
- Evaluating the variance of skill metrics across the dataset.

5. Discussion

This project serves as an effective case study in using data visualization and web technologies to analyze a high-dimensional sports dataset. Several key insights emerge:

- **Player value and performance** are moderately correlated, but not always directly proportional.
- **Wage distribution** is heavily skewed, suggesting a few elite players earn disproportionately more.
- **National trends** indicate that countries often produce players with particular strengths (e.g., Spanish players with high passing scores).
- **Player comparison** via radar charts allows quick assessment of tactical fit.

5.1 Limitations

- The dataset represents simulated player data and may not align with real-world statistics.
- Only static metrics are used; dynamic match-level data would enhance predictive capabilities.
- Goalkeepers and outfield players are not treated separately in radar charts.
- The analysis is restricted to the year 2020 and lacks a temporal progression.

6. Conclusion

This research demonstrates the potential of combining data analysis with modern web technologies to create a dynamic, informative dashboard for sports analytics. The FIFA 21 player dataset, despite being synthetic, offers valuable insights into football player characteristics, valuations, and country-level patterns.

By structuring the dashboard to appeal to multiple user levels, from casual fans to professional analysts, this project exemplifies the accessibility and power of data visualization. Future work can expand this project to include predictive modeling, injury risk analysis, or integration with live match data to provide real-time recommendations.

7. Future Work

- **Incorporate Time-Series Analysis:** Track player development over multiple FIFA editions.
- **Real-World Integration:** Compare with real football data (e.g., from Transfermarkt).
- **Role-Specific Models:** Separate models for goalkeepers, defenders, midfielders, and strikers.
- **User Upload:** Allow users to upload custom player data.

8. References

- EA Sports FIFA 21 Dataset (Kaggle)
- Streamlit Documentation
- Seaborn and Matplotlib Visualization Libraries
- pandas and NumPy Documentation
- "Data Science for Soccer" – Soccerment Blog
- Scikit-learn Documentation (for future ML work)

Appendix

- Full code available in attached Python and Jupyter Notebook files.
- Screenshots of dashboard sections (available upon request).
- CSV format dataset description and dictionary.